

R4 — ML Research Track

Member: _____

Your role

You run the baseline arm (seed 1). You own the official analysis table and the reproducibility analysis.

Weekly tasks

Week 1 — Analyzer notes

- Task: Read `research/tinystories/analyze.py`. Document what the analyzer verifies: cohort tags, expect-arms, expect-seeds, refusal rules.
- Deliverable: The analyzer notes with the week-4 commands.
- Verify: The analyzer command refuses an empty cohort with a clear error.
- Learn: Neural network basics. Write a small MLP in numpy that learns XOR.

Week 2 — Launch baseline seed 1

- Task: Launch the run: `uv run python -m research.tinystories.train --arm baseline --vocab 32768 --steps 3000 --target-core 559000 --seed 1` on your Colab account.
- Deliverable: The run reaches 3000 steps. The checkpoint and the run JSON record land in `runs/`.
- Verify: Checkpoints save to Google Drive every 1000 steps.
- Learn: Embeddings and next-token prediction. Explain what a seed controls in training.

Week 3 — Official analysis table

- Task: When all 6 runs finish, run: `uv run python -m research.tinystories.analyze --tag cleandeploy --expect-arms baseline,ple,fatembed --expect-seeds 2`.
- Deliverable: The preliminary ppl table for all arms and both seeds.
- Verify: The analyzer runs without refusal. The table has 6 rows.
- Learn: The transformer block. Explain why two seeds protect against noise.

Week 4 — Reproducibility analysis

- Task: Write the reproducibility analysis: loss curves vs expectations, threats to validity, dataset drift.
- Deliverable: The reproducibility section of the verification report.
- Verify: R2 quotes one of your threats in the report.
- Learn: Loss and perplexity. Plot the loss curve of your run.

Week 5 — Training loop study

- Task: Study the training loop: learning rate schedule, warmup, batch size. Propose one change.
- Deliverable: The proposal note with expected effect.
- Verify: R3 reviews the proposal before any run.
- Learn: Attention math. Write the QKV attention as numpy code.

Week 6 — Second iteration

- Task: Run the second fine-tune iteration or a second seed of the fine-tune.
- Deliverable: The extra checkpoint and its ppl.
- Verify: The record follows the same format as the first fine-tune.
- Learn: Fine-tuning. Explain why more data helps more than more steps.

Week 7 — Teach KV cache

- Task: Present the KV cache to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: R5 can explain why the cache grows with context.
- Learn: Prepare your presentation deeply.

Week 8 — Presentation materials

- Task: Build the club presentation materials: timeline, gate results, demo flow.
- Deliverable: The presentation deck in the club Drive.
- Verify: R2 reviews the deck before the showcase.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: <https://github.com/slvDev/esp32-ai>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- TinyStories paper: <https://arxiv.org/abs/2305.07759>